# Joshua Nee

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## Education

**Carnegie Mellon University** - *Master of Science in Software Engineering* Aug 2025 – Dec 2026
- GPA: 4.0 / 4.0; Research Assistant: Distributed Systems; Teaching Assistant: Computer Systems, Deep Learning

**National University of Singapore** - *Bachelor of Engineering in Computer Engineering* Aug 2021 – Jan 2025
- GPA: 3.9 / 4.0; Teaching Assistant: Data Structures & Algorithms, Software Engineering & OOP, Machine Learning

## Professional Experience

**DoorDash** May 2026 – Aug 2026
*Software Engineering Intern* *Sunnyvale, CA*
- Shipped the **first DashPass game backend** one week early, serving 2,000 weekly players through Go services, gRPC/Protobuf APIs, Kotlin server-driven UI contracts, and Cassandra storage
- Architected the **backend foundation for all future DashPass games**, adopted by 5 games, with reusable execution graphs, game-agnostic components, and storage interfaces, requiring just 22 lines of per-game registration code
- Enabled 2 engineers new to the codebase to independently build game backends in 3 days each, versus 6 weeks for the original, by authoring an AI skill that codified the system's architecture, code templates, and implementation steps
- Diagnosed 4 silent production defects through distributed traces, uncovering subscription-check inconsistencies during platform migration and resolving win-write failures across iOS and Android without a client release

**Shopee** Jan 2025 – Jul 2025
*Backend Software Engineer* *Singapore*
- Adapted **cross-border order synchronization** for a new resell model with fulfillment checks and conditional order creation in Go within a MySQL-backed e-commerce and payment system processing 10M+ daily transactions
- Reduced directly managed dependencies by 90% across 7 Go services by migrating to a library management framework and restructuring service registration, RPC integrations, and Kafka consumers
- Restored notification-service deployments in 2 weeks for 100+ internal users as the **service's primary contact**, correcting deployment incompatibilities and repairing its failing test environment
- Orchestrated campaign-specific order flows supporting nearly 100,000 free flag distributions for Singapore National Day through configurable shop/item access controls and notification email links, extensible to future campaigns

**ST Engineering** May 2024 – Aug 2024
*Software Engineering Intern* *Singapore*
- Prototyped modular Apple Vision Pro components in Swift for 3D interactions and live geolocation tracking, **reverse-engineering** undocumented visionOS behavior to shorten integration time by 30%

**InTune Auto** Dec 2022 – Dec 2023
*Full-Stack Software Engineer* *New York, NY*
- Developed a scheduler for customer self-booking and automated SMS/email confirmations based on shop availability, among 9+ features from a 4-engineer startup later **acquired by Highland Creek Partners**
- Cut data retrieval latency by 80% on tables with 10K+ rows by redesigning query architecture and implementing keyset pagination in Java, jOOQ, and PostgreSQL for application list views

## Project Experience

**A New Agent Hackathon** - *LLM Agents* Oct 2025 – Oct 2025
- Won the Vapi Track for a tool-using LLM scheduling agent with text and voice dialogue, integrating ClinicalTrials.gov data, Google Calendar, and Supabase Edge to coordinate appointments

**Computer Engineering Capstone Project** - *Machine Learning* Aug 2024 – Nov 2024
- Achieved >95% gesture recognition accuracy using a TensorFlow multilayer perceptron and SHAP-guided features, integrating game subsystems via AES-secured TCP/IP and MQTT with packet validation

## Technical Skills

**Languages & Frameworks**: Python, Go, Java, TypeScript, C++, Bash, React, Next.js, gRPC, Protobuf
**Databases & Communications**: PostgreSQL, MySQL, MongoDB, Firebase, Redis, WebSocket, MQTT, Apache Kafka
**Cloud & DevOps**: AWS (EC2, S3, Lambda), Docker, Kubernetes, NGINX, Grafana, Prometheus
**AI/LLM**: TensorFlow, PyTorch, JAX, LangChain, LangGraph, MCP, RAG pipelines